

considering various factors. Moreover, the proposed method is compared with several data-driven methods using the datasets. Finally, Section V presents the conclusion and future works.

PROTECTION

Convolutional neural network (CNN) is a type of deep neural network, which is especially suitable for classification applications [29]. The aim of training a CNN is to obtain the weight kernels and biases between layers [30]. CNN has fewer parameters to be estimated due to the weight sharing technique. Additionally, it can extract the hidden structure and inherent features more effectively [31]. However, CNNs are burdened with computational complexity [32].

A. CT saturation detection and compensation

B. Percentage Differential Protection

- An accelerated CNN structure is designed by product quantization technique to improve the basic CNN in terms of speed.
- The need for human-expert knowledge and feature engineering process is eliminated because the proposed CNN can discover and extract the effective features of the original signal automatically.
- The proposed method performs robustly in presence of external factors such as CT saturation, series capacitor compensation or superconducting fault current limiter (SFCL) which can make the internal fault detection more difficult.
- The proposed method has high accuracy and outperforms another key and state-of-the-art methods.

$$\overbrace{I_P - I_S}^{\text{Differential Current}} \geq S_D \cdot \underbrace{\frac{I_P + I_S}{2}}_{\text{Restraint Current}} \quad (1)$$

III. ACCELERATED CNN

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exponent. This is a huge disadvantage when dealing with high nonlinearity, and can significantly increase both training and evaluation times. To tackle this issue, CNN is integrated with ReLU activation function [34] and batch normalization [35] to prevent overfitting and vanishing gradient problem and enhance the performance of CNN-based method. Despite this issue, CNN computational cost still remains high for online discrimination of internal fault from inrush phenomena. To tackle this problem, accelerated CNN as a fast and compact version of CNN is developed in this paper. The accelerated CNN performs based on minimizing the approximation error and decomposing the parameters of individual convolution layers and fully-connected layers. A large share of the computational burden is spent on convolutional layers and fully-connected networks (FCN) parameters. In [32], minimizing the approximation error of the individual layer's output is used to compact and accelerate CNN. In this process, convolutional layers and FCN are accelerated by quantizing the weighting matrices in convolutional layers and FCN, respectively.

In this section, the developed accelerated CNN method and different parts of the differential protection procedure are described.

A. Input Data Preparation

To perform fault detection, the input of the developed accelerated CNN ought to be prepared at first. To this end, firstly, a half-cycle of each sample is extracted, and normalized as follows:

$$I_{D-norm} = \frac{I_P - I_S}{\max(|I_P - I_S|)} \quad (2)$$

where I_{D-norm} is the normalized differential phase current.

Then, input normalized dataset includes X_i that is formed as:

$$X_i = \begin{bmatrix} I_{a,D-norm}^1 & I_{a,D-norm}^2 & \cdots \\ I_{b,D-norm}^1 & I_{b,D-norm}^2 & \cdots \\ I_{c,D-norm}^1 & I_{c,D-norm}^2 & \cdots \\ I_{a,D-norm}^s & \cdots & I_{a,D-norm}^N \\ I_{b,D-norm}^s & \cdots & I_{b,D-norm}^N \\ I_{c,D-norm}^s & \cdots & I_{c,D-norm}^N \end{bmatrix}^T \quad (3)$$

$I_{a,D-norm}^s$, $I_{b,D-norm}^s$, and $I_{c,D-norm}^s$ indicate the normalized differential current of phase a , b , and c , respectively and $s = 1, 2, \dots, N$. $N = 56$ is the number of samples in a half cycle [36]. One of the outputs is the binary set $Y^F = (y_j^F) \in \mathbb{B}^{1 \times 4}$ where $F = \{L-G, L-L, LLL, P-S\}$. L-G, L-L, LLL, and P-S stands for line to ground, line to line, three phase, and primary to secondary faults, respectively. The second output is the binary set $Y^D = (y_j^D) \in \mathbb{B}^{1 \times 3}$, where $D = \{Inrush, External\ fault, Over-excitation\}$. The input data are revealed in an accelerated CNN network which outputs the $Y = (y_j) \in \mathbb{B}^{1 \times 1}$ and y_j is defined as follows:

$$y_j = \begin{cases} 1 & \text{if } y_j^F \text{ is detected} \\ 0 & \text{if } y_j^D \text{ is detected} \end{cases} \quad (4)$$

The network is trained by a set of training data $\{X_{(r)}, Y_{(r)}^F, Y_{(r)}^D\} \forall r = 1, \dots, R$.

B. Accelerated Convolutional Layers

In principle, the convolution layer is a feed-forward neural network which projects the low-level maps with local features into high-level maps with global features. The m^{th} feature map at L^{th} layer with activation function f , $y(m, L)$ and $B(m, L)$ bias matrix are:

$$y(m, L) = f\{[W \otimes x(n, L-1)] + B(m, L)\} \quad (5)$$

$$\forall m = 1, \dots, y_{wi} \quad \forall n = 1, \dots, y_{he}$$

$$y_{wi} = \frac{x_{wi} - W_{wi}}{F} + 1 \quad (6)$$

$$y_{he} = \frac{x_{he} - W_{he}}{F} + 1 \quad (7)$$

where F , W , and $x(n, L-1)$ represent the size of the filters, a kernel with learned weight matrix, and n^{th} input map from the previous layer, respectively. Subscript wi and he denotes the width and height. Note that $\otimes$ shows a convolution operation.

The key idea to accelerate layers is using a product quantization technique to decompose feature maps to multiple subspaces. In each subspace, the learning process is performed on a subcodebook. In other words, input data is represented by a subcodebook (concatenation of subcodewords) to enhance the computational efficiency of the CNN. Based on product quantization, each $y(m, L)$ can be quantized as [32]:

$$y(m, L) \approx \sum_{k=1}^K [B(k, L)D(k, L)].S(k, L) \quad (8)$$

$D(k, L)$ and $K(k = 1, \dots, K)$ are the subcodebook and corresponding subcodewords of L^{th} layer, respectively, while, $S(k, L)$ and $B(k, L)$ are defined as input feature map and corresponding bias matrix to each subspace. Each subspace is quantized as (9) by minimizing the following objective:

$$\min_{D(k, L), \{B_p(k, L)\}} \sum_{p(k)} \|D(k, L)B_p(k, L) - W_{p_k}(k)\|_F^2 \quad (9)$$

$$s.t. D(k, L) \in \mathbb{R}^{C_x \times k}, B_p(k, L) \in \{0, 1\}^{k \times C_x'}$$

where $W_{p_k}(k)$ is the original weight vector of the K^{th} subvector at spatial p_k and C_x is the channel number of the feature maps that is decomposed into $K(k = 1, 2, \dots, K)$ subspace ($C_x' = \frac{C_x}{K}$).

However, minimizing (9) may not lead to optimal accelerated convolutional layer, because, optimal quantization error cannot necessarily lead to optimal convolutional layer in terms of fault diagnosis performance. Also, the quantization process is independent of the approximation of each layer. Hence, minimizing the quantization error might cause a large error in obtaining the desirable convolutional layer parameters. To prevent these problems, the approximation error is directly minimized based on the following objective function:

$$\min_{D(k,L), \{B_p(k,L)\}} \sum_{n,p(k)} \|y^*(k,n,L) - \sum_{p(k)} \sum_{k=1}^K (D(k,L)B_p(k,L))^T S_{p_k}(k,n,L)\|_F^2 \quad (10)$$

where $y^*(m,k,L)$ represent the desirable m^{th} output map at L^{th} layer and k^{th} subcodeword.

To obtain optimal solution, the residual error of the k^{th} subspace for the n^{th} differential current as the 2D-input is given as:

$$R(n,k) = y^*(n,k) - \sum_{k' \neq k} [D(n,k')B(n,k')]^T S(n,k') \quad (11)$$

In order to quantize and optimize each subcodebook in each subspace, the overall residual error should be minimized:

$$\min_{D(k,L), \{B_p(k,L)\}} \sum_{n,p(k)} \left\| \sum_{p_k} (D(k,L)B_p(k,L))^T - R_{p_k}(k,n,L) \right\|_F^2 \quad (12)$$

To solve this problem, the following two-step procedure is carried out:

Step1: Consider $B_p(k,L)$ is fixed and then each v^{th} subcodeword is updated sequentially. Firstly, the corresponding residual error of each subcodeword is calculated as:

$$Q_{p_k}(n,v,k,L) = R_{p_k}(n,v,k,L) - \sum_{p_k} \sum_{v' \neq v} [D(v',k,L)]^T S(n,k) \quad (13)$$

and $D(k,L)$ is updated as a least square problem:

$$\min_{D(k,L)} \sum_{n,p(k)} \left\| \sum_{p_k} [D(v,k,L)]^T S(n,k) - Q_{p_k}(n,v,k,L) \right\|_F^2 \quad (14)$$

Step2: $D(k,L)$ is fixed on the found optimal values. Then for finding optimal values of $B_p(k,L)$, residual error for the spatial position p_k is calculated as:

$$P_{p_k}(n,v,k,L) = R_{p_k}(n,v,k,L) - \sum_{p_k' \neq p_k} [D(k,L)B_{p_k'}(k,L)]^T S(n,k) \quad (15)$$

Optimal values for $B_p(k,L)$ are obtained by solving the following problem by exhaustively examining all possibilities:

$$\min_{B_p(k,L)} \sum_{n,p(k)} \left\| \sum_{p_k} [D(k,L)B_p(k,L)]^T S(n,K) - P_{p_k}(n,v,k,L) \right\|_F^2 \quad (16)$$

The optimizing procedure for finding subspace and corresponding parameters is summarized in Algorithm 1.

Algorithm 1 Optimization procedure of accelerated convolutional layers of each subcodebook

- 1: **Step 1: Optimal values for $D(k,L)$**
- 2: Consider $B_p(k,L)$ fixed
- 3: Calculate the corresponding residual error, $Q_{p_k}(n,v,k,L)$ from (13).
- 4: Optimize the problem in (14) as least square problem.
- 5: **Step 2: Optimal values for $B_p(k,L)$**
- 6: Replace $D(k,L)$ values from step 1.
- 7: Calculate the corresponding residual error, $P_{p_k}(n,v,k,L)$ from (15).
- 8: Optimize the problem in (16) by exhaustively examining all possibilities:
- 9: **End**

C. Max Pooling

Max pooling is a nonlinear subsampling layer to pool feature map from the previous layer. Max pooling resolves overfitting in the training and improves computation time by reducing information redundancy, and size. The reason for the size reduction is the overlapped convolution windows [37]. The m^{th} output map at L^{th} layer of the max pooling layer is

$$y(m,L) = f[\omega(m,L).P(x(m,L-1))]\forall m = 1, \dots, y_{wi} \quad (17)$$

where $P(\cdot)$ represents the pooling function and bias matrix in the max-pooling layers.

D. Accelerated FCN

In the last layer of the accelerated CNN, the accelerated FCN is added to connect all hidden states and control the dimension of the accelerated CNN output. The m^{th} output map at L^{th} layer passes through FCN layers to construct the final output map as:

$$y(m,L) = W_f f[W.y(m,L) + B_f(m,L)] \quad (18)$$

where W_f and B_f represent the weight and bias matrices of the FCN.

A procedure similar to accelerated convolutional layers are carried out to speed up the FCN. However, the input of the convolutional layer is 2-D, while the input of the FCN is determined by 1-D samples. To accelerate FCN, the objective function is determined based on the corresponding residual error of k^{th} subspace for n^{th} input data, as:

$$\min_{D(k,L), B_f(k,L)} \sum_n \left\| (R(k,n,L) - D(k,L)B_f(k,L))^T \right\|_F^2 \quad (19)$$

A two-step procedure similar to accelerated convolutional layers is performed to obtain optimal values for $D(k,L)$ and $B_f(k,L)$.

Step1: $B_f(k,L)$ is fixed and subcodebook and corresponding subcodewords are optimized by solving a least square problem in (20):

$$\min_{D(k,L)} \sum_n \|(R(k,n,L) - [D(k,L)]^T S(n,k,L))\|_F^2 \quad (20)$$

Step2: By checking all possible values for subcodewords and pasting obtained values for $D(k,L)$ and selecting the smallest error, (21) that is independent of (20) is solved:

$$\min_{B_f(k,L)} \sum_n \|(R(k,n,L) - [B_f(k,L)]^T [D(k,L)]^T S(n,k,L))\|_F^2 \quad (21)$$

Fig. 1 visualizes the procedure of the accelerated CNN.

E. Overall Structure

In the beginning, the model is fed with normalized differential current as input data. Then, the input data is transformed from 1D to 2D signal and to enhance the learning capability; 2D signal is converted to a $7 \times 8 \times 1$ (rows $\times$ cols $\times$ channels) signal. The 1D samples are not sufficient for complete feature extraction. Therefore, the representation learning technique is applied in which the initial dataset, which is extensively assessed using real input, is converted to 2D-vectors that depends on the sample size. To convert the 1D signal to 2D, the size of the signal should be determined as 7×8 to satisfy $56 = 7 \times 8$ for differential current half-cycle sample. Sequentially, the first eight samples from 56 samples in half-cycle of the input data are arranged as the first row of 2D signal. In the next step, the signal changes to $500 \times (5 \times 6)$ (500 filters with 5×6 kernel size) after it passes the accelerated convolution layer. The number of subcodewords and channels are considered 16 and 8, respectively, to speed up the convolution layer. Then the max pooling layer pools the feature map from the previous layer to resolve overfitting in training and improper computation time. In order to mitigate the vanishing gradient problem as well as enhance the accuracy and speed in the performance of the accelerated CNN, ReLU activation function and batch normalization technique are integrated to accelerated CNN. Interested readers may refer to [34] and [35] for the detailed information on ReLU and batch normalization, respectively. The signal flattens to a $s \times 500$ (sample filters in the previous layer) signal by a flatten layer and then passes through three successive fully connected layers. Consequently, the internal fault is determined in the last fully connected layer based on sigmoid activation function and turns the output into 0 and 1. If the output is 1, the differential relay detects the internal fault and sends a trip signal. Note that, except in the last FCN layer that sigmoid activation function is used, activation function in the designed accelerated CNN is ReLU. Besides, the filters stride is set (1, 1) and zero padding is used in the accelerated CNN.

IV. RESULTS

In this section, five cases are discussed to address the advantages of the proposed differential protection scheme. The cases are: 1) simple benchmark, 2) CT saturation caused by fault or inrush current, 3) power transformer with series

capacitor compensation, 4) transformer with SFCL on neutral point, and 5) experimental setup. In each case changes in fault type, fault location, switching angle, transformer winding's connection, and source impedance are taken into account. The following indexes are used to evaluate the results [38].

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \% \quad (22)$$

$$DEP = \frac{TN}{TN + FP} \% \quad (23)$$

$$SEC = \frac{TP}{FN + TP} \% \quad (24)$$

$$SAF = \frac{TP}{TP + FP} \% \quad (25)$$

In (22) to (25) TN Shows the number of detected faults; FP is the number of not detected faults; TP is the number of non-fault conditions that do not activate the differential relay, and FN is the number of healthy conditions that are mistakenly diagnosed as fault and activated the differential relay [38].

The *accuracy* (ACC) shows the authenticity of the fault detection method which is defined as the number of correctly detected disturbances over the total number of detections, including correct ones and false ones (22). The *dependability* (DEP) demonstrates the ability of the fault detection method to detect and isolate the internal faults. It is defined as the ratio of the number of detected internal faults over the number of all internal faults used during the testing stage (23). The *security* (SEC) index illustrates the ratio of detected inrush current and external faults to the actual number of these two disturbances which is defined as (24). The *safety* (SAF) defined in (25) evaluates the capability of the proposed method to isolate internal faults from other disturbances. In other words, it demonstrates the ability of a fault diagnosis method where inrush current and external faults are not mistakenly detected as internal faults.

PSCAD/EMTDC software package is used to simulate a slightly modified part of a 230 kV network with four power transformers (Fig. 2). Sampling frequency is 112 samples per cycle. The generated data is further processed in Keras package [39] which is a high-level neural networks application programming interface written in Python. Keras is capable of working on top of TensorFlow or Theano. It is developed to enable fast experimentation. Keras allows easy and fast prototyping on a computer. The data are processed on a 3.4 GHz processor, 32 GB memory, a 64-bit OS, and a GeForce GTX 1080 TI graphical processing unit (GPU). In all cases, the samples are randomly partitioned into a training and testing set with 70% to 30% ratio, respectively. The testing stage is performed by K-fold cross validation [40].

To prove the performance of the proposed accelerated CNN in preventing overfitting problems, Fig. 3 shows the accuracy versus epoch, i.e., a complete pass through a given dataset. This figure verifies that the fault diagnosis is not overfitting.

In order to visualize the fault diagnosis result in the differential protection of the power transformers, we used a t-distributional stochastic neighbor embedding (t-SNE) [41] of

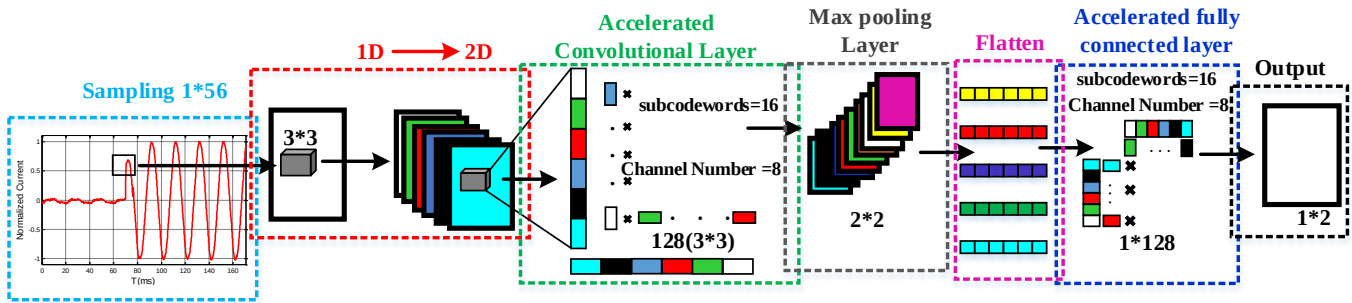


Fig. 1: Procedure of the proposed differential protection scheme of the power transformers

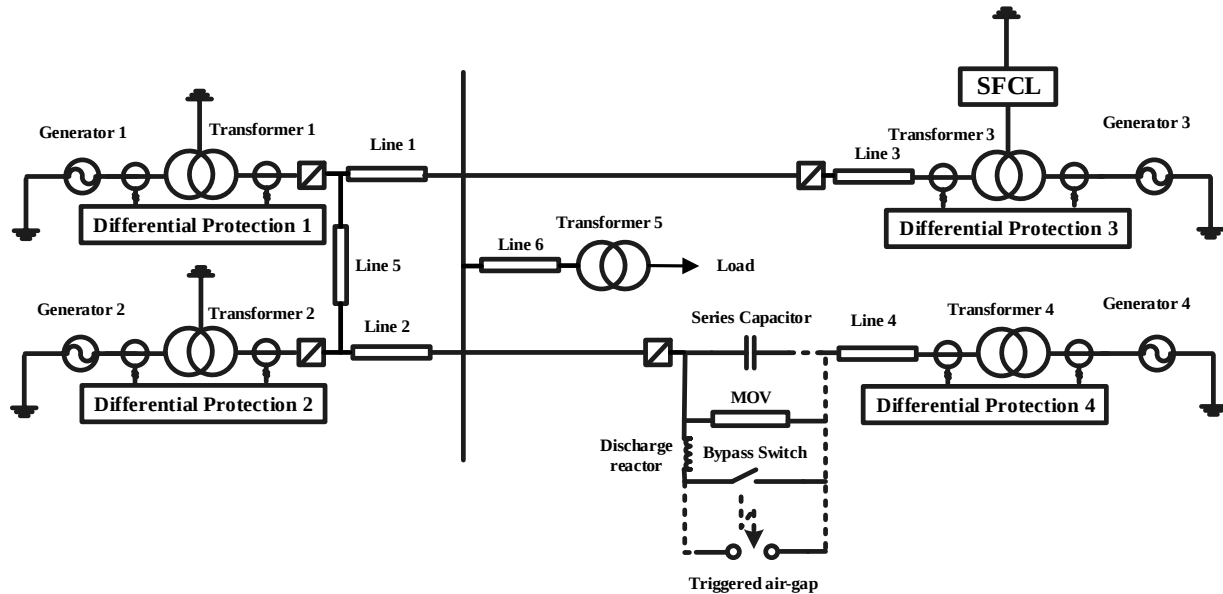


Fig. 2: Single line diagram of a modified part of the 230 kV power network

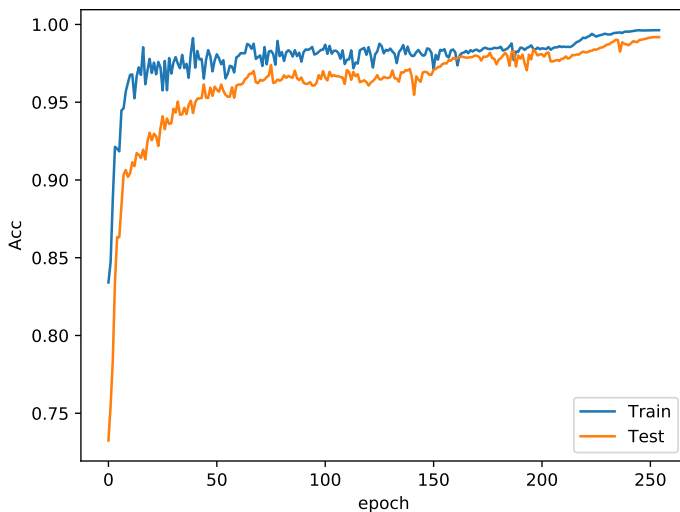


Fig. 3: Accuracy vs. epoch in the fault diagnosis

fused features illustrated on 2D-feature maps in Fig. 4 in case 2. Fig. 4a shows that t-SNE is unable to diagnose internal faults from other non-fault disturbances. Figure 4b shows the

significant impression on feature learning in the proposed method training process.

1) *Case 1:* For the first case, 4488 samples are generated for fault current, inrush current and external fault excluding any external factors. Fault currents are generated based on different fault types, winding connections, source impedances, fault inception angles, and fault locations. Inrush currents are generated with various residual fluxes, winding connections, source impedances, and switching instant fault inception angles are taken into account. External fault data set contains eight fault types, 12 fault inception angles and four impedance values.

The numerical results of the proposed method in detecting four internal faults, inrush current and external faults is represented in table. I. The results illustrate that all indexes are more than 98.46% fault detection, while for inrush current and external faults the method operates completely correct. In over excitation conditions, no change is observed in performance of the proposed method, and it performs completely correct.

2) *Case 2:* In this case, the effect of CT saturation is taken into account. Decay DC component of short circuit fault and large DC component of inrush current are two major reasons of CT saturation [33]. A reliable differential protection should

TABLE III: Performance of Proposed accelerated CNN in Case 3

Cases	LG Fault	LL Fault	LLL Fault	P-S Fault	Inrush	External Fault	Over Excitation
ACC%	99.64	99.52	100	99.52	100	100	99.42
DEP%	98.46	98.46	100	98.46	100	100	99.23
SEC%	99.86	99.71	100	99.72	100	100	99.50
SAF%	99.71	99.71	100	99.72	100	100	99.67

current within the first cycle may distort the input signals of the differential protection. As a result, the differential relay cannot diagnose the fault to send the trip signal. This problem can also happen during an inrush current if it exceeds the critical current of the SFCL. The SFCL model is added to the transformer 3 in Fig. 2. Parameters of SFCL are set based on the procedure in [36].

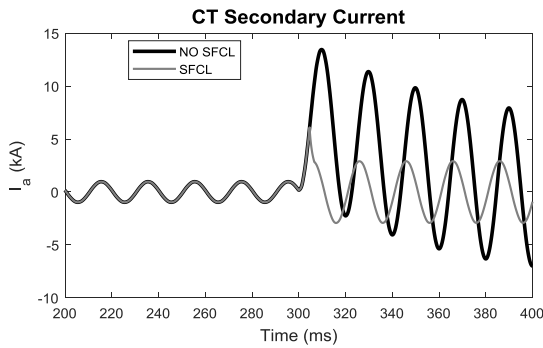


Fig. 6: Internal fault current in solidly grounded and SFCL grounded neutral point

A sample waveform of internal fault current in solidly grounded and SFCL grounded neutral point is represented in Fig 6.

The dataset of this case is identical to case 1. Table. IV represents the performance indexes for this case.

TABLE IV: Performance of Proposed accelerated CNN in Case 4

Cases	LG Fault	LL Fault	LLL Fault	P-S Fault	Inrush	External Fault	Over Excitation
ACC%	99.52	99.28	100	98.91	99.33	99.85	99.65
DEP%	97.70	98.46	100	96.92	99.31	99.11	99.61
SEC%	99.57	99.43	100	99.28	99.35	99.92	99.67
SAF%	99.86	99.71	100	99.43	99.48	99.92	99.83

5) *Case 5*: This case includes an experimental test bench to practically evaluate the proposed method. The experimental data is gathered using a 1 kVA, 50 Hz, and 380/380 V three phase transformer, shown in Fig. 7. Several points inside the windings are accessible through terminals which are used to create internal faults. 0.6-3kV, 2.5VA CTs are used for current measurement. The accuracy class of the CTs is 3. 4488 cases were carried out, and measurements were recorded by a digital storage oscilloscope with 112 samples per cycle rate. The rated values of experimental devices are lower than devices

in transmission scale due to the limits we faced in selecting the devices. However, the proposed approach used normalized data, so we can compare the simulated and experimental results.



Fig. 7: Experimental test system

Table. V shows the performance of the proposed method using experimental results. As we can see, the proposed method provides a very good performance when it is fed with the experimental data. For instance, for LG and LL faults, the reliability indexes are at least 98.46% and for LLL faults DEP and SEC indexes are 100%.

TABLE V: Performance of Proposed accelerated CNN in Case 5

Cases	LG Fault	LL Fault	LLL Fault	P-S Fault	Inrush	External Fault
ACC%	99.15	99.28	99.64	99.28	99.26	99.55
DEP%	98.46	98.46	100	97.69	99.31	99.11
SEC%	99.28	99.43	100	99.57	99.48	99.92
SAF%	99.71	99.71	99.57	99.57	99.22	99.59

V. COMPARING DIFFERENT METHODS

In this section, the results obtained in each case are compared with several methods. Four state-of-the-art deep architecture methods including gated recurrent unit (GRU), CNN, long short-term memory (LSTM), deep belief Boltzmann machine (DBM) are considered. Furthermore, four shallow data-driven methods such as SVM, KNN, LVQ, ANN, and two well-known signal processing methods including restricted second harmonic and gradient-vector methods are considered. These methods are briefly described as follows:

- GRU is implemented with ReLU activation function, 256 epochs, and 128 unit, and GRU consists of three GRU layers and two fully-connected layers.
- CNN is utilized with 20 feature maps, 128 filter, 256 epochs, and maximum pooling size of three, with sigmoid activation function.
- LSTM is implemented with ReLU activation function, 256 epochs, and 128 unit and LSTM consists of three LSTM layers and two fully-connected layers.
- DBM performs with 86 neurons in input layer, four hidden layers with 150 neurons in each of the layers, one neuron in the output layer, and the *sigmoid* activation function.

- SVM performs with radial basis function (RBF) kernel and cross validation.
- KNN performs with one nearest neighbor based on Euclidean distance and Baye's discussion rule.
- LVQ performs based on Euclidean distance, with six input and six hidden layers with corresponding rate of one and two neurons in output layer.
- ANN performs with 86 neurons in input layer, two parallel hidden layers with 128 and 64 neurons, and one neuron in the output layer.
- Restricted second harmonic is applied based on [9].
- Gradient-vector method is applied according to [8].

Table VI presents a comparison between all methods in the described cases except case 1 and Table VII compares the average performance time of all deep and shallow based methods.

In addition to reliability and accuracy index, receiver operating characteristics (ROC) curve and area under the ROC curve (AUC) [43] are used to compare the performance of all methods. ROC curve shows the false positive rate (FPR) and true positive rate (TPR) on x-axis and y-axis, respectively. ROC and AUC can provide comprehensive evaluation based on a relative trade-off between FRP and TPR. Figure 8 demonstrates the effectiveness of the proposed differential protection scheme.

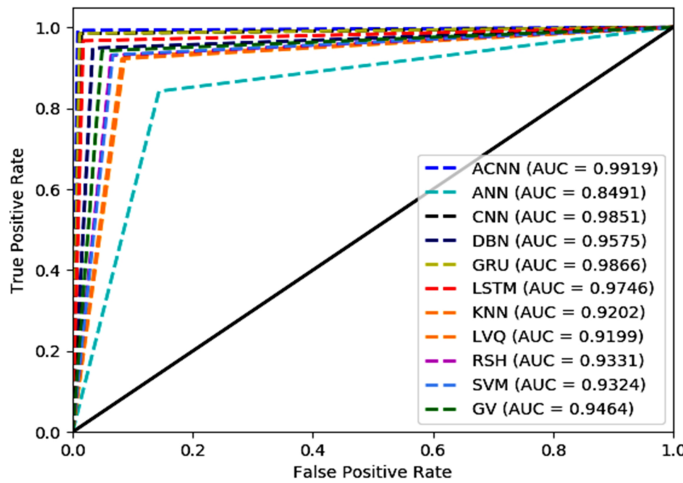


Fig. 8: ROC curve in case 2

A. The Proposed CNN vs. Shallow Data-Driven Methods

The accelerated CNN based method is compared with four data-driven methods with shallow architecture, including SVM, KNN, LVQ, and ANN.

The difference between the performances of shallow architecture methods and deep ones can easily be noticed in table VI where the performances of the three deep architecture methods are obviously better. For instance, accelerated CNN performs 5.94% and 7.16% more accurately than SVM and KNN in case 3, respectively. We can also observe this in Fig 8. Among the shallow architecture methods, ANN has the least reliable performance. KNN, LVQ, and SVM have almost similar performances except in the second case where SVM provides

a better performance. As it can be seen in Table VII, the data-driven methods with shallow architecture perform faster than the proposed method. However, the proposed method is more reliable and performs significantly more accurate, and its computational complexity is suitable for real-time application.

B. The Proposed CNN vs. Deep Data-Driven Methods

Accelerated CNN has a slightly better performance than the conventional CNN in all cases. For instance, in Case 5 accelerated CNN performs 0.14% more securely than basic CNN. In most of the cases, the proposed accelerated CNN has a slightly better performance than GRU, which is also a state-of-the-art deep learning method. What makes accelerated CNN a better choice is the faster performance it provides compared to conventional CNN, GRU, LSTM, and DBM methods. Based on Table VII, basic CNN, GRU, LSTM, and DBM perform fault detection in 13.56 ms, 11.2 ms, 14.48 ms, and 16.56 ms, respectively, while accelerated CNN performs in 4.12 ms, and it is three to five times faster than other methods. Furthermore, accelerated CNN is the only deep learning method that can be implemented in the real-time application. Because accelerated CNN detects the internal faults much faster than the measurement time (a half cycle=10ms). Thus, it can be entitled as a real-time fault detection method.

C. The Proposed CNN vs. Signal Processing Methods

According to the numbers presented in table VI the proposed accelerated CNN is significantly more reliable than restricted second harmonic and gradient-vector. Especially, in case 4 where the signals are distorted by SFCL as an external factor, the restricted second harmonic method experiences a severe decrease in the reliability. In this case, the reliability indexes are between 48.92% and 59.31%. It is important to notice that restricted second harmonic method is used in most of the real differential relays in power transformers.

VI. CONCLUSION

The deficiencies of traditional methods in differential protection to distinguish internal faults from the inrush current in power transformers, such as computational burden and dependency on the model and predefined threshold, decrease the reliability and generality of this crucial process. The proposed method makes a key step toward practical intelligent automation of power transformers' protection benefiting from the vast amount of data recorded by the modern data acquisition systems. We designed an accelerated CNN by using product quantization technique to speed up the convolution and FCN layers. The accelerated CNN performs four times faster than basic CNN not only without losing accuracy but also with about 1% accuracy improvement. Moreover, the proposed machine learning based protection method can be applied to different systems regardless of the system parameters once the CNN structure is set. The proposed method was applied to a system in both simulated and experimental cases. The results of accelerated CNN method were compared to ten other methods, including eight data-driven and two signal processing methods. The numerical results validate the quality of the proposed method.

TABLE VI: Reliability Criteria Comparison Between the Proposed Accelerated CNN and Shallow and Deep Structure Classifiers

	Case 2				Case 3				Case 4				Case 5			
	ACC %	DEP %	SAF %	SEC %	ACC %	DEP %	SAF %	SEC %	ACC %	DEP %	SAF %	SEC %	ACC %	DEP %	SAF %	SEC %
Accelerated CNN	99.35	99.17	99.61	98.85	99.18	99.08	99.28	99.14	98.81	98.62	99.00	98.71	99.26	99.23	99.28	99.28
GRU	99.03	98.89	99.22	98.46	98.66	98.77	98.57	98.85	98.59	98.31	98.85	98.43	99.41	99.38	99.43	99.43
CNN	98.06	98.61	97.29	98.05	98.52	98.46	98.57	98.57	98.44	98.15	98.71	98.29	99.04	99.08	99.00	99.14
LSTM	98.06	98.89	98.43	96.90	97.46	98.15	98.26	96.85	96.94	97.34	97.54	96.56	98.37	98.62	98.71	98.14
DBN	93.37	96.11	94.29	89.53	95.75	96.46	96.65	95.13	95.10	95.23	95.53	94.99	96.15	96.00	96.28	96.28
SVM [13]	91.75	93.89	88.76	91.24	93.24	92.77	93.71	93.31	91.79	89.91	93.57	90.72	93.63	93.69	93.57	94.11
KNN [15]	90.42	92.46	87.60	89.33	92.02	90.58	93.41	91.32	91.91	90.77	92.98	91.54	92.63	92.26	92.98	92.85
LVQ [14]	89.48	91.39	86.82	87.84	91.99	91.08	92.84	91.78	91.54	90.46	92.55	91.24	93.03	92.00	93.98	92.66
ANN [5]	89.84	91.11	87.45	86.43	84.91	84.62	85.24	85.61	84.12	83.23	84.69	84.47	84.87	84.77	84.69	85.69
Restricted second harmonic [9]	84.79	86.67	82.17	81.54	93.31	93.08	93.59	93.29	54.30	48.92	59.31	55.50	93.76	93.38	94.12	93.85
Gradient vector [8]	91.10	93.33	87.98	90.44	94.64	94.77	94.59	95.13	81.23	80.46	81.95	81.83	90.10	83.40	97.42	84.28

TABLE VII: Comparison of Average Performance Time of the Deep and Shallow Based Methods

Methods	Accelerated CNN	GRU	CNN	LSTM	DBM	SVM	KNN	LVQ	ANN
Test time (ms)	4.12	11.2	13.56	14.48	16.56	2.46	2.42	2.56	3.86

Despite the highly accurate, reliable, and fast performance of the proposed method, this method requires large dataset in the training process that may be difficult to collect or generate. Therefore, we have to utilize artificial data generation methods to enrich the collected dataset which needs further investigations in the future.

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VII. APPENDIX

Table VIII illustrates the parameters of the test system in Fig. 2.

TABLE VIII: Parameters of Simulated System

Component	Specifications	Component	Specification
Gen. 1	$V_n = 15.75kV$ $S_n = 115MVA$	Tran. 5	$V_n = 400/115$ $/31.5kV$ $S_n = 700MVA$
Gen. 2	$V_n = 21kV$ $S_n = 1455MVA$	Line 1	$42.7km$ $Z_1 = 1.3 + j14.6\Omega$ $Z_0 = 10.7 + j43.7\Omega$
Gen. 3	$V_n = 15.75kV$ $S_n = 1140MVA$	Line 2	$82.9km$ $Z_1 = 2.8 + j28.3\Omega$ $Z_0 = 23.5 + j77.4\Omega$
Gen. 4	$V_n = 15.75kV$ $S_n = 370MVA$	Line 3	$15.8km$ $Z_1 = 0.46 + j5.4\Omega$ $Z_0 = 4.4 + j14.3\Omega$
Tran. 1	$V_n = 15.75/400kV$ $S_n = 115MVA$	Line 4	$35.1km$ $Z_1 = 1.0 + j12.0\Omega$ $Z_0 = 9.8 + j31.9\Omega$
Tran. 2	$V_n = 21/400kV$ $S_n = 1450MVA$	Line 5	$98.7km$ $Z_1 = 2.9 + j33.7\Omega$ $Z_0 = 27.5 + j89.6\Omega$
Tran. 3	$V_n = 15.75/400kV$ $S_n = 1190MVA$	Line 6	$40.3km$ $Z_1 = 1.2 + j13.8\Omega$ $Z_0 = 11.2 + j36.6\Omega$
Tran. 4	$V_n = 15.75/400kV$ $S_n = 370MVA$	Load	$V_n = 115kV$ $S_n = 700MVA$ $PF = 0.95$